INCH-POUND

MIL-B-43404F
16 August 1993
SUPERSEDING
MIL-B-43404E
29 June 1984

MILITARY SPECIFICATION

BEEF STEW, COOKED, DEHYDRATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dehydrated, cooked beef stew for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

A-A-1898	-	Cushioning Material, Cellulosic, Packaging
L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	-	Aluminum Foil
PPP-B-26	-	Bag, Plastic (General Purpose)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-C-1752 - Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible
PPP-C-1797 - Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam
PPP-T-60 - Tape: Packaging, Waterproof

MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence Items, General Specification For

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of military and federal specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Institutional Meat Purchase Specifications for Fresh Beef, Series 100

(Copies are available from the Chief, Livestock and Meat Standardization Branch, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2603, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

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Meat and Poultry Inspection Regulations (9 CFR Part 301-391)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

United States Standards for Condition of Food Containers

(Copies are available from the Chairperson, Condition of Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. Standards for Grades of Carrots for Processing (7 CFR Part 51)

U.S. Standards for Grades of Fresh Shelled Peas for Canning or Freezing (7 CFR Part 51)

(Copies are available from the Head, Standardization Section, Fresh Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2056, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

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(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, One Maritime Plaza, 23rd Floor, San Francisco, CA 94111.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by Density Gradient Technique

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP) and the National Formulary (NF)

(Application for copies should be addressed to the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

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3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any one or any combination of the following USDA certified Institutional Meat Purchase Specification (IMPS) cuts: Item No. 114 (shoulder clod), Item No. 161 (round, shank-off, boneless), Item No. 163 (round, shank-off, 3-way, boneless), Item No. 165 (round, rump and shank-off, boneless), Item No. 167A (round, knuckle, peeled), Item No. 169 (top round), Item No. 170A (bottom round, heel out), and Item No. 171B (outside round). All beef shall be certified by a USDA Agricultural Marketing Service (AMS) agent for condition of the product and compliance with the Institutional Meat Purchase Specifications Quality Assurance Provisions.

NOTE: Certified IMPS cuts may be further trimmed (see 3.3.1) and broken down into smaller cuts after certification as requested by the purchaser. Boxes used for shipment of the further trimmed and broken down cuts shall be labeled, "Beef for further processing as specified in MIL-B-43404". All beef fat trimmings shall be included with the shipment. The certificate shall be noted as such and shall accompany the shipment.

3.2.1.1 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product (see 3.3.1) shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained in the temperature range of 28°F to 50°F, inclusive.

b. Holding in the fresh-chilled state for not more than four days after initial certification is permitted, provided that the beef is maintained in the temperature range of 28°F to 40°F, inclusive.

c. Holding in the frozen state for not more than 180 days after initial certification is permitted, provided that the beef is:

- frozen to 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence

d. Tempering/thawing of frozen beef shall comply with the following:

- microwave tempering/thawing of frozen beef is permitted provided excellent condition of the beef, as determined by USDA, is maintained

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- the microwave tempering/thawing process shall not exceed 48 hours from time of removal of the beef from the freezer to further preparation
- regardless of the process used, frozen beef shall be tempered/thawed to a temperature not to exceed 40°F
- when a tempering/thawing process other than microwave is used, the time from the start of the tempering/thawing process (removal from freezer) to further preparation shall not exceed 96 hours
- beef which has been tempered/thawed shall not be refrozen

3.2.1.2 Beef, cooked, frozen. The frozen, cooked beef logs or dices shall be prepared from beef that complies with 3.2.1 and 3.2.1.1. The beef shall be processed in accordance with 3.3.1, 3.3.1.1, and 3.3.1.2 as applicable. The cooked beef logs shall be frozen in their original cooking casings. The cooked, beef logs or dices shall be packaged and sealed in USDA approved packaging material and shall be frozen to an internal temperature of 0°F or below within 72 hours after being placed in the freezer. The initial freezing (in-storage) date and processing and packaging requirements shall be certified by a USDA Agricultural Marketing Service (AMS) Livestock Division agent. The frozen, packaged, cooked beef logs or dices shall have been held at an internal temperature of 0°F or below for a period not to exceed 120 days prior to further processing. Dices prepared from previously frozen logs shall not be frozen.

3.2.2 Potatoes.

3.2.2.1 Potatoes, fresh. The potatoes shall be fresh, firm, clean, sound, and of a white flesh variety. The specific gravity shall be not less than 1.068 or not greater than 1.081 (total solids not less than 18 percent or greater than 21 percent). The reducing sugar content shall be not more than 2.0 percent on a dry weight basis.

3.2.2.2 Potatoes, diced, water blanched, frozen. The frozen, potato dices shall be prepared from fresh potatoes meeting the requirements of 3.2.2.1 and shall be from the latest season's crop. The frozen, potato dices shall be processed, handled, and packaged in a manner so as to prevent discoloration. Frozen potato dices shall be held at 0°F or below until further processing.

3.2.3 Carrots.

3.2.3.1 Carrots, fresh. The fresh carrots shall be U.S. No. 2 or better of the U.S. Standards for Grades of Carrots for Processing.

3.2.3.2 Carrots, diced, individually quick frozen (IQF). The individually quick frozen (IQF) diced carrots shall have been prepared from carrots meeting the requirements of 3.2.3.1 and 3.3.2.3. The IQF carrot dices shall be of the latest season's pack.

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3.2.4 Peas, green.

3.2.4.1 Peas, green, fresh, shelled. The fresh, shelled, green peas shall be U.S. No. 2 or better of the U.S. Standards for Grades of Fresh Shelled Peas for Canning or Freezing. The peas shall be of such size as not to pass through a 9/32 inch sieve. Peas shall be slit or perforated to improve rehydration characteristics.

3.2.4.2 Peas, green, shelled, individually quick frozen (IQF). The individually quick frozen (IQF), shelled, green peas shall have been prepared from fresh, green peas meeting the requirements of 3.2.4.1 and 3.3.2.4. The IQF, shelled, green peas shall be of the latest season's pack. Peas shall be slit or perforated to improve rehydration characteristics.

3.2.5 Water. Water used for formulation and blanching shall conform to the National Primary Drinking Water Regulations.

3.2.6 Oil, corn. Corn oil shall possess a clean, bland flavor, and shall have a minimum stability of 25 hours (A.O.M.), a free fatty acid value not to exceed 0.05 percent, a moisture/volatile matter content not to exceed 0.06 percent and pass a cold test of 5.5 hours (minimum).

3.2.7 Starch, freeze-thaw stable. Freeze-thaw, stable starch shall form a smooth, short textured, heavy bodied, almost clear, aqueous solution. The flavor of the cooked solution shall be bland with essentially no cereal or starchy taste.

3.2.8 Soup stock, dehydrated. Dehydrated soup stock shall be a light brown to tan colored, free-flowing powder derived from soup stock produced from USDA inspected beef bones with adhering meat. The dehydrated soup stock shall have a minimum of 93 percent total solids and a maximum of 2 percent fat. There shall be no added salt in the product.

3.2.9 Onion, chopped, dehydrated. Dehydrated, chopped onions shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.10 Salt. Salt shall be white, refined sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.11 Onion powder. Onion powder shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.12 Caramel color, powder. Powdered caramel color shall comply with the Food Chemicals Codex.

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3.2.13 Pepper, black, ground. Ground, black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic, penetrating odor, a hot, biting, pungent flavor and a light gray to speckled, black-gray color. The ground pepper shall contain not less than 2.0 mL of volatile oil per 100 grams of ground, black pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.14 Garlic powder. Garlic powder shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.15 Celery seed, ground. Ground celery seed shall be derived from the seed of Apium graveolens L. and shall be light to rich-brown in color, and shall possess a characteristic, celery odor and flavor with a warm, slightly bitter aftertaste. The ground celery seed shall contain not less than 2.0 mL of volatile oil per 100 grams and shall be of such size that not less than 95 percent, by weight, shall pass through a U.S. Standard No. 55 sieve.

3.2.16 Bay leaves, ground. Ground bay leaves shall be derived from the dried leaves of Laurus nobilis L., and shall possess a pleasant, aromatic odor and pungent, mildly bitter flavor with a pale green to yellow-green color. The volatile oil content shall be not less than 1.0 mL per 100 grams of ground bay leaves. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 30 sieve.

3.2.17 Allspice, ground. Ground allspice shall be derived from dried, nearly ripe fruit of Pimenta officinalis L. and shall possess a fragrant, clove-like aroma, strongly aromatic, pungent, clove-like flavor and a dark reddish-brown color. Allspice shall contain not less than 3.0 mL of volatile oil per 100 grams of ground allspice with 80.0 percent eugenol as a principal constituent. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 30 sieve.

3.2.18 Nitrogen. Nitrogen shall meet the requirements of the United States Pharmacopoeia and shall be water or liquid nitrogen pumped.

3.2.19 Preblended spice and seasoning mix. Preblended spices and seasonings may be used. The spices and seasonings in the mix shall comply with the individual requirements specified in this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the spice and seasoning mix formula (see 3.3.4).

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. Beef, meeting the requirements of 3.2.1, shall be further trimmed, if necessary, in such a manner as to ensure compliance with

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the finished product requirements (see 3.7). The boned and trimmed beef shall be mechanically, coarse ground through a grinder plate having openings measuring 1/2 inch to 1 inch (inclusive) in diameter. The fat content of the ground beef shall be not less than 8 percent or more than 16 percent (see 4.5.2.1). If the fat content requires adjustment, original cuts and/or clean fat trimmings from the original cuts, shall be chopped to reduce the particle size and shall be added to the lean, coarse ground meat. The chopped portion shall not exceed 25 percent of the total batch size.

3.3.1.1 Beef log preparation and processing. The beef logs shall be prepared and processed as follows:

Ingredients	Percent by weight
Beef	99.0
Salt	1.0

a. The coarse ground beef shall be mechanically vacuum mixed with the salt. The mixing shall continue until the mixture exhibits a sticky (tacky) consistency. Time from grinding to mixing shall not exceed four hours. Temperature of the beef shall not exceed 40°F during this four hour period. Dry ice may be used to control the temperature during mixing.

b. The mixed beef shall be mechanically, tightly stuffed into suitable casings of a size to meet dice requirements (see 3.3.1.2). The beef logs shall be cooked within 24 hours after being stuffed. Total time from coarse grinding to cooking shall not exceed 28 hours. The internal temperature of the logs shall be maintained at 28°F to 40°F prior to cooking.

c. The beef logs shall be cooked in water or steam until an internal temperature of not less than 155°F and not greater than 160°F is reached.

d. Immediately after completion of the cooking process, the cooked beef logs shall be placed in an ice water bath, or cooled by other commercially acceptable cooling methods, and rapidly cooled to 130°F or below. The beef logs shall be further cooled from 130°F to 80°F or below within 1-1/2 hours and further cooled from 80°F to 40°F or below within five hours. The cooled, cooked beef logs shall be held not longer than 48 hours at an internal temperature of 28°F to 40°F after removal from the cooking process and prior to dicing. Alternatively, if the cooked beef logs are to be held for more than 48 hours after cooking, they shall be placed in a freezer within four hours after cooking and frozen to an internal temperature of 0°F or lower within 72 hours after being placed in the freezer (see 3.2.1.2). If the beef logs are to be frozen, the meat shall not have been previously frozen. The frozen, cooked beef logs shall be held at an internal temperature of 0°F or below for a period of not more than 120 days prior to dicing. The frozen, cooked beef logs shall be tempered to an internal temperature of 28°F to 40°F prior to dicing.

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3.3.1.2 Dicing of cooked beef logs. The cooked beef logs shall be mechanically diced to yield uniform dices of approximately 1/2 by 1/2 by 1/4 inch (± 1/16 inch in each dimension). The dices shall be held not longer than 48 hours at an internal temperature of 28°F to 40°F until product preparation or, if they are to be stored in the frozen form, until they are packaged and sealed (see 3.2.1.2). Dices prepared from previously frozen logs shall not be frozen.

3.3.2 Vegetable preparation.

3.3.2.1 Preparation of fresh potatoes. The fresh potatoes shall be thoroughly cleaned to remove dark and extraneous material, peeled, trimmed, and thoroughly washed. The washed potatoes shall be mechanically diced with dicer setting at 3/8 by 3/8 by 3/8 inch (± 1/32 inch in each dimension). The potato dices shall be cooked to assure compliance with finished product requirements. The cooked potatoes shall be immediately cooled with cooling water to the initial temperature of the cooling water and thoroughly drained. The cooled, drained potatoes shall be handled in a manner to prevent discoloration and shall be used in product preparation within four hours after cooking.

3.3.2.2 Frozen, potato dice preparation. Frozen potato dices shall be cooked and handled as specified in 3.3.2.1.

3.3.2.3 Fresh, carrot preparation. The clean, fresh carrots shall be peeled, trimmed, and mechanically diced with dicer settings of 3/8 by 3/8 by 3/8 inch (± 1/32 inch in each dimension). The diced carrots shall be blanched only to the extent necessary to prevent discoloration and to remove excess air. The blanched carrots shall be immediately cooled with cooling water to the initial temperature of the cooling water and thoroughly drained. The cooled, drained carrots shall be handled in a manner to prevent discoloration and used in product preparation or individually quick frozen within 4 hours after blanching.

3.3.2.4 Fresh, green pea preparation. The fresh, green peas, slit or perforated, shall be blanched only to the extent necessary to prevent discoloration and to remove excess air. The blanched peas shall be immediately cooled with cooling water to the initial temperature of the cooling water and thoroughly drained. The cooled, drained peas shall be handled in a manner to prevent discoloration and shall be used in product preparation or individually quick frozen within 4 hours after blanching.

3.3.3 Frozen, vegetable preparation. Frozen vegetables shall be thawed only to the extent necessary to accommodate product preparation procedures.

3.3.4 Spice and seasoning mix preparation. The spice and seasoning mix shall be a uniform blend of the following ingredients:

Ingredient	Percent by weight
Soup stock, dehydrated	32.0

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Onions, chopped, dehydrated	28.2
Onion powder	18.0
Salt	15.6
Caramel color powder	2.1
Pepper, black, ground	1.7
Garlic powder	0.7
Celery seed, ground	0.7
Bay leaves, ground	0.5
Allspice, ground	0.5

3.3.5 Product preparation. The beef stew shall be formulated and prepared as follows:

Ingredient	Percent by weight
Water	30.0
Beef, cooked, diced	25.0
Potatoes, fresh or frozen, diced	20.0
Peas, slit or perforated	10.0
Carrots, raw, diced	10.0
Spice and seasoning mix	2.25
Corn oil	2.0
Starch	0.75

NOTE: The following product preparation procedures were used in the development of this product. Alternative procedures may be used provided finished product requirements are met. When alternative procedures are used, the time and temperature requirements specified for the prepared product are still applicable.

a. A slurry shall be made using the starch and the water.

b. Add the spice and seasoning mix to the slurry and mix well. Heat to 180°F with continuous mixing.

c. Add the beef dices, peas, carrots, and corn oil and heat to not less than 160°F and not more than 180°F.

d. The cooked potatoes (see 3.3.2.1 or 3.3.2.2) shall be added to the mixture and mixed only to the extent necessary to assure a uniform blend of ingredients.

e. The product shall be spread on dehydrator trays immediately after completion of the cooking process.

3.4 Freezing. The prepared product shall be spread on dehydrator trays and cooled immediately, placed in the freezer within 4 hours after cooking, and frozen to 0°F or below within 6 hours after being placed in the

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freezer. The frozen product shall be maintained at 0°F or lower at all times prior to freeze dehydration and shall be freeze dehydrated within 5 days after freezing.

3.5 Dehydration. The product shall be freeze dehydrated at an absolute pressure not to exceed 1.5 mm of mercury and at a product temperature as indicated by suitable instruments, not to exceed 150°F. If the platen temperature is maintained at 155°F or below with radiant heating, the product temperature may be disregarded. After dehydration is completed, the pressure shall be equalized to atmospheric level with nitrogen and the product shall be packaged as specified in 5.1. In no case shall more than 16 hours elapse between the time the chamber is opened and the time the product is completely packaged. During the interim the product shall be adequately protected from oxygen and moisture by either holding under a nitrogen atmosphere with 2.0 percent or less oxygen, or under a vacuum of at least 27 inches of mercury for the entire period. If a vacuum is used, it shall be broken with nitrogen.

3.6 Pouch and mixing bag filling and sealing. Each pouch and mixing bag shall be filled and sealed with product such as to conform to the finished product requirements and to the requirements specified in 5.1.1.

3.7 Finished product requirements.

3.7.1 Dehydrated product. The dehydrated product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor, such as, but not limited to, burnt, scorched, moldy, rancid, sour, or stale.

c. There shall be no color foreign to the product.

d. There shall be no evidence of incomplete dehydration such as wet or soft spots.

e. There shall be no evidence of faulty dehydration such as glazed surface areas or burnt areas measuring more than 0.5 inch in any dimension or dark colored cores of any dimension.

f. At least 30 percent by weight of the product tested shall be retained on a U.S. Standard No. 4 sieve.

g. The moisture content of the dehydrated product shall not exceed 2.0 percent.

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h. The individual net weight shall be not less than 120 grams for the Ration, Cold Weather (RCW) pouch and shall be not less than 140 grams for the Long Range Patrol (LRP) pouch.

i. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples, and not greater than 150,000 per gram in any sample.

j. The E. coli count shall be less than 3 per gram (no positives in the standard 3 tube MPN technique) in four of five samples and not greater than 10 per gram in any sample.

3.7.2 Rehydrated product. Rehydration shall be in accordance with the specified instructions in 4.5.6. The rehydrated product shall comply with the following requirements:

a. The rehydrated product shall show complete water penetration through all particles. However, gristle and similar materials in the meat do not necessarily rehydrate properly and such materials shall not be considered as evidence of incomplete rehydration.

b. The rehydrated product shall possess flavor, odor, and texture considered normal for frozen and reheated beef stew.

c. The total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, shall be not more than 9 grams per 120 grams (RCW) or 11 grams per 140 grams (LRP) of dehydrated product which has been rehydrated.

d. No individual pouch shall contain a bone piece measuring 0.3 inch or more in any dimension.

3.7.3 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.8 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Products Inspection Regulations of the U.S. Department of Agriculture.

3.9 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of the obligation and responsibility to

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deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the methods described in the Official Methods of Analysis of the AOAC and in the Approved Methods of the American Association of Cereal Chemists (AACC), to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.1.2 Mixing bag, baseboard, and pouch material certification. Material listed below may be accepted on the basis of a contractor's certificate of conformance to the indicated requirements. In addition, compliance to the requirements for inside pouch dimensions and dimensions of manufacturer's seals may be verified by a certificate of conformance.

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Requirement	Requirement paragraph	Test procedure
Mixing bag material identification	5.1.1.1	Laboratory evaluation
Baseboard (when used)	5.1.1.1.2	Laboratory evaluation
Thickness of films for laminated material	5.1.1.2	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378
Aluminum foil thickness	5.1.1.2	As specified in QQ-A-1876
Laminated material identification and construction	5.1.1.2	Laboratory evaluation
Color of laminated material	5.1.1.2	Visual examination

4.5.1.3 Unfilled mixing bag examination. The unfilled mixing bags shall be examined for the defects listed in table I. The lot size shall be expressed in bags. The sample unit shall be one bag. The inspection level shall be S-4 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65 for major defects.

TABLE I. Unfilled mixing bag defects

Category	Defect
Major	
101	Not formed as specified
102	Inside dimensions not as specified
103	Tear, hole, or open seal
104	Bags stick together
105	Not clean
106	Foreign odor

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4.5.1.4 Unfilled mixing bag leakage testing. Eight unfilled mixing bags shall be selected at random from the lot regardless of lot size and shall be tested for leakage as follows: Each sample bag shall be filled with eight ounces of boiling water (being careful not to spill any water on the outside) and suspended 8 to 12 inches above a sheet of clean paper for 10 minutes. Any moisture on the paper attributable to bag leakage shall be classified as a major defect. The lot size shall be expressed in bags. The sample unit shall be one bag. The finding of any defect shall be cause for rejection of the lot.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation, or of other valid documents shall be cause for rejection of the involved product.

4.5.2.1 In-process testing for fat content. Four 1 pound samples shall be selected at random from the day's production of raw, ground beef. Each of the four samples shall be tested for fat content in accordance with the Official Methods of Analysis of the AOAC, method 960.39, 976.21 or 985.15. Results shall be reported to the nearest 0.1 percent. Any result not conforming to the fat content requirement in 3.3.1 shall be classified as a major defect and shall be cause for rejection of the lot of ground beef or any product made therefrom.

4.5.3 Net weight examination.

4.5.3.1 Net weight examination of pouches. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty mixing bag and pouch. Any individual net weight of less than 120 grams for a RCW pouch or less than 140 grams for a LRP pouch shall be classified as minor defect. Results shall be reported to the nearest gram. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5.4 Filled and sealed pouch examination. The filled and sealed pouches shall be examined for the defects listed in table II. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 4.0 for minor defects.

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TABLE II. Filled and sealed pouch defects 1/

Category		Defect
Major	Minor	
101		Tear, cut, hole, or abrasion through one or more layers in the pouch material or leakage through any heat seal 2/
102		Foldover wrinkle extending into the seal such that the effective closure seal is reduced to less than 1/16 inch 3/
103		Evidence of entrapped product that reduces the effective closure seal to less than 1/16 inch 3/
104		Presence of delamination 4/
105		Unclean pouch 5/
106		Labeling is missing, incorrect, or illegible
107		Any impression or design on the heat seal surfaces which conceals or impairs visual detection of seal defects 6/
108		Less than 3/16 inch between inside edge of tear notch and inside edge of seal
109		Closure seal width not as specified
110		Closure seal not located as specified
	201	Tear notch(es) or serrations missing or not located as specified
	202	Depth of tear notch(es) or serrations not as specified
	203	Presence of delamination 4/

1/ Any evidence of rodent or insect infestation shall be cause for rejection of the lot.

2/ Pinholes or breaks inherent to the manufacturing process of the aluminum foil shall not be scored as a defect.

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3/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/16 inch wide, from side seal to side seal that produces a hermetically sealed pouch.

4/ Delamination defect classification:

Major - Delamination of the outer ply in the pouch seal area that can be propagated to expose aluminum foil at the food product edge of the pouch after manual flexing of the delaminated area. To flex, the delaminated area shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delaminated area shall then be rapidly flexed 10 times by rotating both hands in alternating clockwise, counter-clockwise directions. Care shall be exercised when flexing delaminated areas near the tear notches to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between thumb and forefinger and gently lifted toward the food product edge of the seal or if the separated area is too small to be held between thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to the product edge of the seal with no discernible resistance to the gentle lifting, the delamination shall be scored as a major defect. Additionally, spot delamination of the outer ply in the body of the pouch that is able to be propagated beyond its initial borders is a major defect. To determine if the delaminated area is a defect, use the following procedure. Mark the outside edges of the delaminated area using a bold, permanent marking pen. Open the pouch and remove the contents. Cut the pouch transversely not closer than 1/4 inch (± 1/16 inch) from the delaminated area. The pouch shall be flexed in the area in question using the procedure described above. Any propa- gation of the delaminated area, as evidenced by the delaminated area exceeding the limits of the outlined borders, shall be scored as a major defect.

Minor - Minor delamination of the outer ply in the pouch seal area is acceptable and shall not be classified as a minor defect unless it extends to within 1/16 inch of the food product edge of the seal. All other minor outer ply delamination in the pouch seal area or isolated spots of delamination in the body of the pouch that do not propagate when flexed as described above shall be classified as minor defects.

5/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal filings, etc.) or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which represents no health hazard or potential pouch damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

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b. Dried product which affects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

6/ If doubt exists as to whether or not the sealing equipment leaves an impression or design on the closure seal surface that could conceal or impair visual detection of seal defects, samples shall be furnished to the contracting officer for a determination as to acceptability.

4.5.5 Pouch vacuum examination. The filled and sealed pouches shall be visually examined for conformance to the vacuum requirement in 5.1.1.3 not less than 96 hours after filling and sealing. The sealed pouch shall continue to exhibit tight adherence to the surface contours of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between thumb and forefinger of each hand, while simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.6 Product examination. The finished product shall be examined for the defects listed in table III. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE III. Product defects 1/ 2/

Category		Defect
Major	Minor	Applicable to dehydrated product
101		Evidence of incomplete dehydration such as wet or soft spots
102		Presence of glazed surface areas measuring more than 0.5 inch in any dimension
103		Presence of burned areas as indicated by a black spot greater than 0.5 inch in any dimension
104		Presence of dark colored cores in any dimension
	201	Less than 30 percent of product retained on a U.S. Standard No. 4 sieve 3/

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TABLE III. Product defects 1/ 2/

Category		Defect
Major	Minor	Applicable to rehydrated product 4/
105		Incomplete water penetration through all particles 5/
106		Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, is more than 9 grams per 120 grams (RCW) or 11 grams per 140 grams (LRP) of dehydrated product which has been rehydrated
107		Presence of a bone piece measuring 0.3 inch or more in any dimension
108		Product does not possess flavor, odor, and texture considered normal for frozen and reheated beef stew

1/ The presence of any foreign material (for example, glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (for example burnt, scorched, moldy, rancid, sour, stale, etc.), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability or overall appearance shall be cause for rejection of the lot (see 3.7.3).

3/ The contents of two randomly selected, 120 gram or 140 gram pouches as applicable, shall be mechanically shaken for two minutes for sieve testing.

4/ Multiply the product weight in grams by 2.9 to get milliliters of hot water for rehydration. Continue with label instructions.

5/ Dry areas attributable to gristle and similar material shall not be considered as defects.

4.5.7 Internal pressure test. Eight filled and sealed pouches shall be selected at random from the lot regardless of lot size and individually tested for internal pressure resistance. The internal pressure resistance shall be determined by pressurizing the pouches while they are restrained between two rigid plates spaced 1/2 inch (± 1/16 inch) apart. If a three-seal tester (one that pressurizes the pouch through an open end) is used, the closure seal shall be cut off for testing of the side and bottom

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seals of the pouch; for testing of the closure seal, the bottom seal shall be cut off. The pouches shall be emptied prior to testing. If a four-seal tester (designed to pressurize filled pouches by use of a hypodermic needle through the pouch wall) is used, all four seals can be tested simultaneously. The distance between the rigid restraining plates on the four seal tester, shall be equal to the thickness of the product (+ 1/16 inch). Pressure shall be applied at the approximate uniform rate of 1 psig per second until 14 psig pressure is reached. The 14 psig pressure shall be held constant for 30 seconds and then released. The pouches shall then be examined for separation or yield of the heat seals. Any rupture of the pouch or evidence of seal separation greater than 1/16 inch in the pouch manufacturers seals shall be considered a test failure. Any seal separation that reduces the effective closure seal width to less than 1/16 inch (see Table II) shall be considered a test failure. Any test failure shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.8 Moisture content testing. Eight filled and sealed pouches shall be selected at random from the lot regardless of the lot size. The contents of each pouch shall be blended to uniformity using a blender or a food processor. The blending must be rapid and conducted in such a way that minimum heat is transferred to the product and that the product has minimum exposure to atmospheric moisture. Each sample shall be individually tested for moisture content in accordance with the Official Methods of Analysis of the AOAC, method 950.46A, except that the temperature-time cycle shall be modified by using a temperature of 70°C for 16 hours at a pressure of not more than 100 mm of mercury. Test results shall be reported to the nearest 0.1 percent. Any result not conforming to the moisture content requirement in 3.7.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.9 Microbiological testing. Five filled and sealed pouches shall be selected at random from the lot regardless of the lot size. Each sample shall be individually tested for aerobic plate count and E. coli in accordance with the Official Methods of Analysis of the AOAC, methods 966.23 and 966.24. The diluent shall be added to each sample of dry product and allowed to stand for 15 minutes before the blending of the sample. Continue with the procedure as directed. Any result not conforming to the microbiological requirements in 3.7.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.10 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major:	Marking missing, incorrect, or illegible
	Cushioning material missing or not as specified
Minor:	More than 40 pounds packed in box
	Pouches not packed flat in layers

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4.5.11 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. The product shall be filled into mixing bags specified in 5.1.1.1 and sealed in pouches as specified in 5.1.1.2.

5.1.1.1 Mixing bags. A net weight of 120 grams of product for the RCW or 140 grams of product for the LRP shall be filled into mixing bags fabricated from 0.003 inch thick polyethylene made in accordance with type I bags of PPP-B-26, except that the bag shall be made from food grade polyethylene. The bags shall not inpart any undesirable odor or flavor to the product and shall be approved by the FDA for food use. The inside dimensions of the bag shall be 5-1/2 by 3-1/2 by 8 inches (± 1/8 inch in each dimension) with the open end of the bag having the 5-1/2 by 3-1/2 inch dimensions. After filling, the side panels of the bag (corresponding to the 3-1/2 inch dimension) shall be folded in to form gussets, and the bag shall be closed by folding over its open end. Where required by the bulkiness of the product, the bulk of the contents within the bag shall be compressed such that the product thickness is not less than 2-1/4 inches by applying a constant uniform pressure over the entire top surface of the bag. The open top of the filled bag shall be folded over to prevent spillage of the product. Alternatively, polyethylene bags conforming to type II of PPP-B-26, with inside dimensions of 9 inches by 8 inches (± 1/8 inch in each dimension) with the open end of the bag having the 9 inch dimension, may be used. The material shall be as specified for type I bags. When type II bags are used, they shall be formed over a baseboard on a mandrel as specified in 5.1.1.1.1 before filling. While the bag is positioned on the mandrel, the two bag ears shall be taped down over a 5-1/2 by 3-1/2 inch baseboard constructed in accordance with 5.1.1.1.2. An eight inch long strip of 1 inch wide tape complying with type III, class 2 of PPP-T-60, shall be applied across both ears to fasten them securely to the baseboard. Alternatively, both type I and type II mixing bags may be fabricated from unpigmented 0.003 inch thick high density 0.941 to 0.965 g/cm^3 food grade, flat polyethylene tubing. type I and type II mixing bags, fabricated from clear polyethylene or the alternate material, shall be of adequate strength and integrity to withstand the addition of boiling water when tested in accordance with 4.5.1.4.

5.1.1.1.1 Mandrel. The mandrel face shall be 5-1/8 by 3-3/8 inches and the height shall be not less than 12 inches. All corners and edges of the mandrel shall be rounded and all surfaces finished to prevent damage to the bag material.

5.1.1.1.2 Baseboard. The baseboard shall be constructed from unbleached and uncoated solid sulfate or sulfite paperboard not less than 0.022 inch thick. Alternatively, the baseboard may be constructed from 0.022 inch thick, food grade paperboard made from recycled fibers. If paperboard made

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from recycled fibers is used, it shall be natural color ranging from brown to grey. The baseboard shall be 5-1/2 by 3-1/2 inches (± 1/8 inch in each dimension) and shall have all corners rounded to form a smooth contour.

5.1.1.2 Barrier pouches. The pouch shall be constructed from a multiple ply, flexible laminate, barrier material consisting of an outer ply of 0.0005 inch thick polyester bonded to an intermediate ply of 0.0007 inch thick aluminum foil with 10 pounds per ream pigmented polyethylene, and bonding the opposite side of the aluminum foil to 0.003 inch thick ionomer (inner ply) or a blend of not less than 50 percent linear, low density polyethylene and polyethylene. The linear, low density polyethylene portion of the blend shall be the copolymer of ethylene and octene-1, having a melt index range of 0.8 to 1.2 g/10 minutes in accordance with ASTM D 1238 and a density range of 0.918 to 0.922 g/cm^3 in accordance with ASTM D 1505. For LRP applications, the complete exterior surface of the pouch shall be uniformly colored in the range of green 34079 through 34087 or 24052 through 24087 or brown 30045 through 30118 (excluding 30109) or 10045 of FED-STD-595. For RCW applications, the complete, exterior surface of the pouch shall be colored white overall with a color in the range of 37778 through 37886 of FED-STD-595. All tolerances for thickness of pouch materials shall be plus or minus 20 percent. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being packaged. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches.

5.1.1.2.1 Pouch construction. The pouch shall be a flat style, preformed pouch having inside dimensions of 6 inches (± 1/8 inch) by 9 inches (± 1/8 inch). The first dimension specified shall be the opening of the pouch between the heat sealed sides. The empty pouch shall be made by heat sealing three edges with 3/8 inch (± 1/8 inch) wide seals. A "V", "C", or block "U" tear notch at least 1/32 inch deep, located 1 to 1-1/4 inches from the top edge of the pouch, shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the pouch may be provided with an extended or foldover lip, extended not more than 5/16 inch to facilitate opening and filling. Tear notch location shall be measured from the top of the pouch, excluding the extended or foldover lip. Alternatively, if the pouch has serrated edges, the serrations may be used as tear notches provided that the serrations are sharp (no plastic tailings exist) and the serrations depth and the minimum seal width at the serrated edges are in accordance with the above requirements. The pouch shall be constructed with square or rounded corners at all four corners (radius approximately 3/8 inch).

5.1.1.3 Pouch filling and sealing. The filled mixing bag specified in 5.1.1.1 shall be inserted into a pouch as specified in 5.1.1.2 in such a manner to avoid puncturing the pouch material or contaminating the heat seal area. The filled pouch shall be sealed under a vacuum level of 23 inches of mercury. The sealed pouch shall show no loss of vacuum when examined as

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specified in 4.5.5. The filled pouch shall have a 3/8 inch (± 1/8 inch) heat seal. If a thermal impulse closure seal is used, a seal width of 1/8 inch to 7/16 inch will be acceptable. When tested as specified in 4.5.7, the filled and sealed pouch shall withstand an internal pressure of 14 psig for 30 seconds without rupture or seal separation greater than 1/16 inch or seal separation that reduces the effective closure seal width to less than 1/16 inch. The closure seal shall not extend below the tear notch on either side of the pouch. The closure seal shall be free of fold over wrinkles which reduce the effective closure seal to less than 1/16 inch.

5.2 Packing. Packing shall be level C.

5.2.1 Level C packing. Not more than 40 pounds of pouched product shall be packed flat in a fiberboard box constructed and closed in accordance with style RSC-L, class - domestic, variety SW, grade 200 of PPP-B-636. Cushioning material not less than 1/8 inch thick, conforming to grade II, class B or C of A-A-1898 or cushioning material not less than 1/16 inch thick, conforming to type I of PPP-C-1797 or cushioning material conforming to type 7, class 1 of PPP-C-1752 shall be placed between each of the layers and in the bottom and top of the box to minimize movement of the individual packages.

5.3 Unit loads. When specified (see 6.1), the product packed as specified in 5.2.1 shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling and marking.

5.4.1 Pouches. Each pouch shall be clearly printed or stamped, in a manner that does not damage the pouch, with permanent black ink which is free of carcinogenic elements or ingredients. The information shall be located on the body of the pouch not closer than 1/16 inch to any seal. If a non-contact type printer is used, the information may be located anywhere on the pouch (in one complete print), except the closure seal area. The label shall contain the following information:

BEEF STEW, COOKED, DEHYDRATED
(letters not less than 7/32 inch high)
ADD 12-14 OUNCES OF HOT WATER (APPROXIMATELY 1/2 CANTEEN CUP) TO MIXING BAG. STIR, WAIT APPROXIMATELY 5 MINUTES.
HOT WATER MAY BE ADDED IN STAGES TO KEEP FOOD HOT.
(letters not less than 1/8 inch high)

Ingredients
(letters not less than 3/32 inch high)
Date 1/
(letters not less than 1/8 inch high)
Net weight
(letters not less than 1/8 inch high)

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Official establishment number (for example, EST-38/USDA stamp)
(letters not less than 1/8 inch high)
Contractor's name and address
(letters not less than 1/8 inch high)

1/ Each pouch shall have the date of pack noted by using a four digit code beginning with the final digit of the current year followed by the three digit Julian day code. For example, March 19, 1991 would read 1079. The Julian day code shall represent the date the product is unit packed into the pouch.

5.4.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.3 Unit load marking. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.7.3 and 6.2).
e. Type and class of unit load when loading is required (see 5.3).
f. Ration, Cold Weather or Long Range Patrol component.

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and number of units to be furnished. The contracting officer also should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Subject term (key word) listing.

Dehydrated food
Long Range Patrol
Pouched food
Ration, Cold Weather
Shelf stable food

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6.4 Changes from previous issue. Marginal notations are not used in the revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

Preparing activity:

Army - GL

(Project 8940-0740)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-43404F	16 August 1993

3. DOCUMENT TITLE

BEEF STEW, COOKED, DEHYDRATED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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